

4.5

$$\text{Matrix } A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\lambda I = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \dots A - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

Eigenvalues of Matrix A
is 2 & 5.

$$\det A = |A - \lambda I| = (4-\lambda)(3-\lambda) - (1)(2)$$

$$= 12 - 4\lambda - 3\lambda + \lambda^2 - 2$$

$$= \lambda^2 - 7\lambda + 10$$

$$(\lambda - 2)(\lambda - 5)$$

$$\text{Matrix } B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \quad |B - \lambda I| = 0 \dots \begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 4 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

$$\det B = (1-\lambda) \begin{bmatrix} 1-\lambda & 4 \\ 0 & 1-\lambda \end{bmatrix} - (2) \begin{bmatrix} 0 & 4 \\ 0 & 1-\lambda \end{bmatrix} + (3) \begin{bmatrix} 0 & 1-\lambda \\ 0 & 0 \end{bmatrix}$$

$$= (1-\lambda) [(1-\lambda)(1-\lambda) - (4)(0)] + (-2) [(0)(1-\lambda) - (4)(0)] + (3) [(0)(0) - (1-\lambda)(0)]$$

$$\downarrow$$

$$(1-\lambda)(1-\lambda)^2 + (-2)(0) + (3)(0)$$

$$(1-\lambda)(\lambda^2 - 2\lambda + 1)$$

$$\lambda^2 - 2\lambda + 1 - \lambda^3 + 2\lambda^2 - \lambda$$

$$-\lambda^3 + 3\lambda^2 - 3\lambda + 1 = 0 \dots \lambda = 1$$

Eigenvalues of Matrix B is 1

$$\lambda = 1 \left| \begin{array}{cccc} -1 & 3 & -3 & 1 \\ & -1 & 2 & -1 \\ & & -1 & 2 & -1 & 0 \end{array} \right|$$

$$-\lambda^2 + 2\lambda - 1$$

$$\lambda = 1$$

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \dots \quad |C - \lambda I| = 0 \dots \begin{bmatrix} 1-\lambda & 2 & 3 \\ 4 & 5-\lambda & 6 \\ 7 & 8 & 9-\lambda \end{bmatrix}$$

$$\begin{aligned} \det C &= (1-\lambda)[(5-\lambda)(9-\lambda) - (6)(8)] - (2)[(4)(9-\lambda) - (6)(7)] + (3)[(4)(8) - (5-\lambda)(7)] \\ &= (1-\lambda)[(5-\lambda)(9-\lambda) - 48] - (2)[36 - 4\lambda - 42] + (3)[32 - 35 + 7\lambda] \\ &= (1-\lambda)[45 - 5\lambda - 9\lambda + \lambda^2 - 48] - (2)[-4\lambda - 6] + (3)[-3 + 7\lambda] \\ &= (1-\lambda)[\lambda^2 - 14\lambda - 3] + 8\lambda + 12 - 9 + 21\lambda \\ &= \lambda^2 - 14\lambda - 3 - \lambda^3 + 14\lambda^2 + 8\lambda + 3 + 21\lambda \\ &= -\lambda^3 + 15\lambda^2 + 18\lambda \dots = 0 \quad \lambda = 0, \quad \frac{15}{2} \pm \frac{3\sqrt{33}}{2} \end{aligned}$$

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix} \quad \dots \quad |D - \lambda I| = 0 \dots \begin{bmatrix} 1-\lambda & 2 & 3 \\ 4 & 5-\lambda & 6 \\ 7 & 8 & 10-\lambda \end{bmatrix}$$

$$\begin{aligned} \det D &= (1-\lambda)[(5-\lambda)(10-\lambda)] - (2)[(4)(10-\lambda) - 42] - 9 + 21\lambda \\ &= (1-\lambda)[50 - 15\lambda + \lambda^2] - (2)[-4\lambda - 2] - 9 + 21\lambda \\ &= 50 - 15\lambda + \lambda^2 - 50\lambda + 15\lambda^2 - \lambda^3 + 8\lambda + 4 - 9 + 21\lambda \\ &= -\lambda^3 + 16\lambda^2 + 12\lambda - 3 \end{aligned}$$

$$\lambda = 16.7075, 0.1983, -0.9057$$